Agreement Analysis: sPCI vs pPCI

Date: April 16, 2026
Source: `notebooks/01-agreement_all.ipynb`

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1. Data Overview

- **Sample size:** n = 454 paired measurements
- Two continuous variables: `sPCI` and `pPCI`

Statistic	sPCI	pPCI
Mean	15.872	8.930
SD	9.469	6.219
Skewness	0.504	0.494

Both distributions are mildly right-skewed. sPCI shows systematically higher values than pPCI.

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2. Systematic Bias

The mean difference (sPCI − pPCI) is **6.943**, and the median difference is **5.0**, indicating a large and consistent positive bias of sPCI over pPCI across all observations.

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3. Hypothesis Tests for Systematic Difference

All tests strongly reject the null hypothesis of no systematic difference:

Test	Statistic	p-value
Paired permutation test (mean)	6.943	< 0.0001
Paired permutation test (median)	5.0	< 0.0001
Sign test	399 / 429 positive signs	1.96×10^{-83}

The sign test found 399 out of 429 non-tied pairs where sPCI > pPCI, confirming that the bias is overwhelmingly directional.

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4. Confidence Intervals for the Difference

Mean difference

Method	Lower	Upper
Percentile bootstrap (20,000 resamples)	6.313	7.586

Method	Lower	Upper
BCa bootstrap	6.325	7.599

Median difference

Method	Lower	Upper
Percentile bootstrap	4.0	6.0
BCa bootstrap	4.0	5.0

The BCa CIs are narrow and entirely above zero, confirming a robust, non-trivial systematic offset.

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5. Bland-Altman Analysis

Classical Bland-Altman

Parameter	Value
Mean bias (sPCI − pPCI)	**6.943**
SD of differences	6.861
Upper limit of agreement (+1.96 SD)	**20.39**
Lower limit of agreement (−1.96 SD)	**−6.50**

The limits of agreement span nearly 27 units (−6.5 to 20.4), indicating very wide variability between the two methods beyond the systematic offset.

Robust Bland-Altman (quantile-based)

Parameter	Value
Median bias	5.000
Robust LoA (2.5th – 97.5th percentile)	[−2.0, 25.0]

Proportional Bias

A significant proportional bias was detected (regression of differences on means):

Parameter	Value
Slope	0.486
Intercept	0.913
Pearson r	0.514
p-value	6.12×10^{-32}

The positive slope indicates that the discrepancy between sPCI and pPCI **grows as the average value increases** — methods diverge more at higher PCI levels.

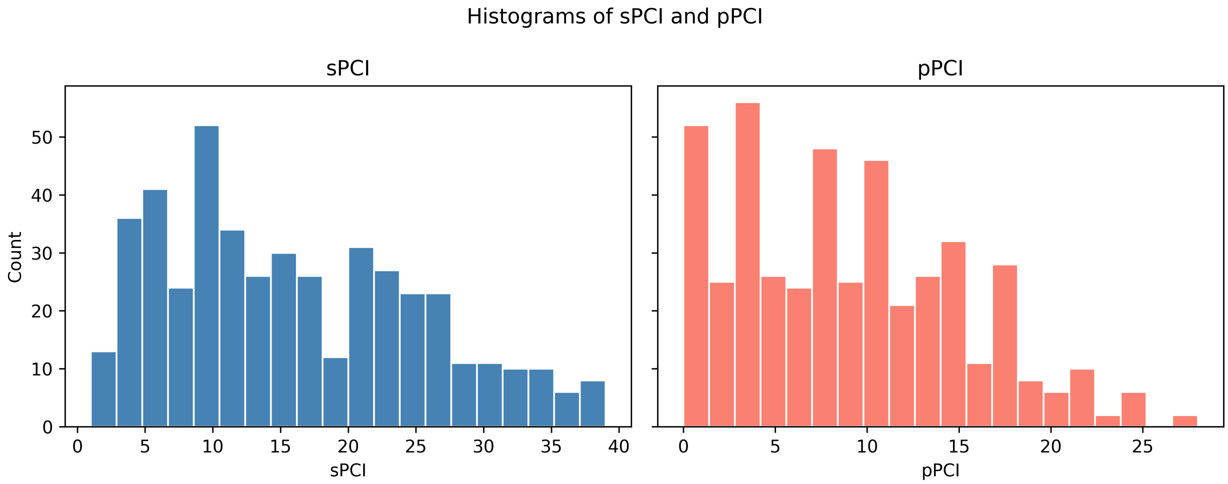

Figure 1: Hisotgrams

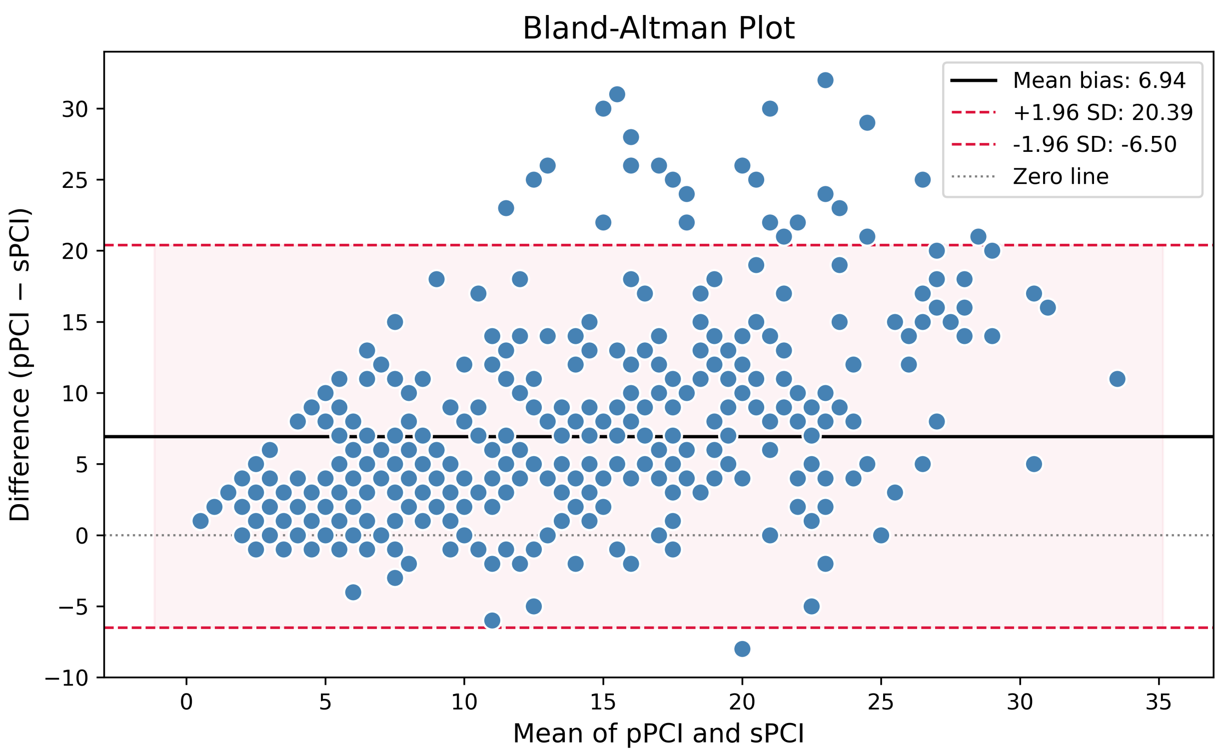

Figure 2: Bland-Altman Plot

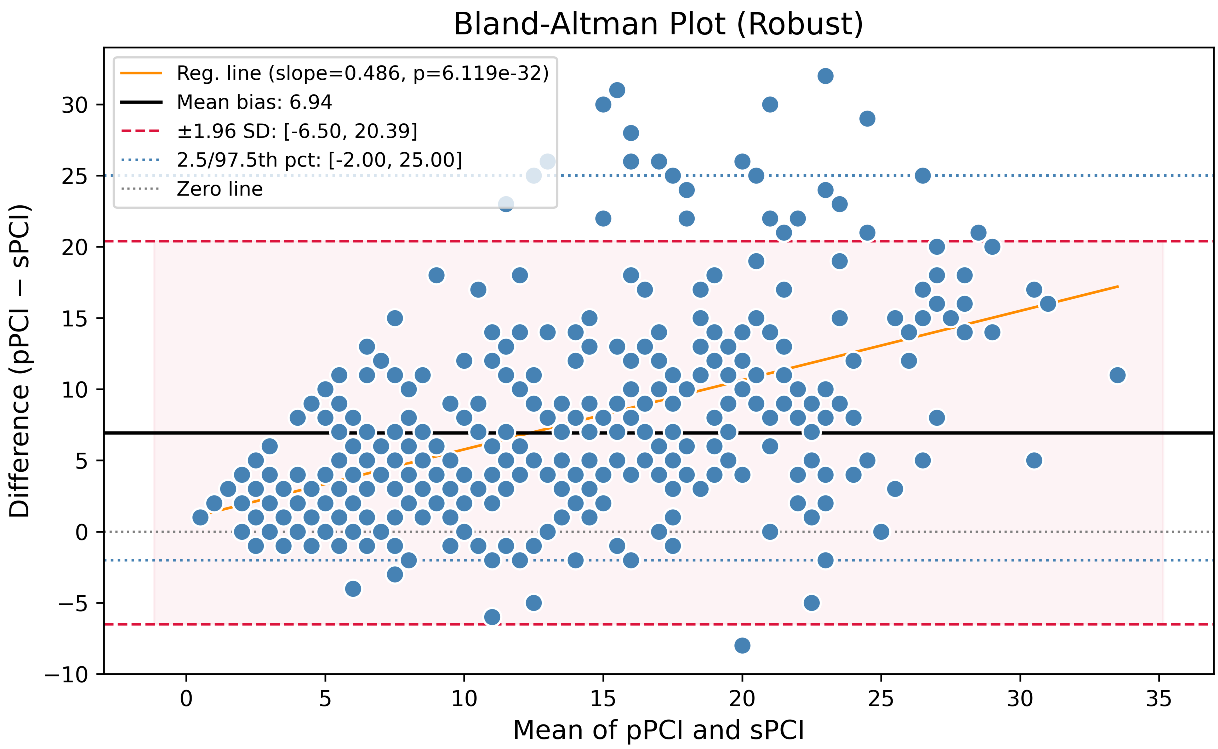

Figure 3: Bland-Altman Plot

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6. Intraclass Correlation (ICC)

Using ICC(A,1) (absolute agreement, single rater):

Without bias correction

Model	ICC	95% CI	p-value
ICC(A,1)	**0.461**	[−0.0, 0.70]	< 0.0001

The ICC of 0.46 reflects only moderate absolute agreement, heavily penalised by the large systematic bias.

After bias correction (median shift = 5.0 removed from sPCI)

Model	ICC	95% CI	p-value
ICC(A,1)	**0.616**	[0.54, 0.68]	7.74×10^{-53}

After removing the median bias, ICC improves to 0.62, indicating **moderate-to-good consistency** in the relative ordering of values between methods, but with still substantial residual variability.

7. Summary and Interpretation

sPCI and pPCI **cannot be considered interchangeable**:

1. **Large systematic bias:** sPCI exceeds pPCI by approximately 5–7 units on average (depending on mean vs median), which is clinically meaningful given the range of 0–40.
2. **Wide limits of agreement:** Even ignoring the bias, differences between methods range from roughly -6.5 to $+20.4$ (classical) or -2 to $+25$ (robust), making individual-level substitution unreliable.
3. **Proportional bias:** The disagreement is worse for patients with higher PCI values, which is particularly concerning if the two methods are applied selectively by severity.
4. **Moderate consistency after bias correction:** An ICC of 0.62 suggests the methods track the same underlying construct reasonably well in rank terms, but the fixed and proportional offsets remain a barrier to direct comparison.

Recommendation: If both methods are used in the same study, a recalibration or harmonisation step is necessary before combining or directly comparing sPCI and pPCI scores.